

Lab 2 Practice

This practice does not need to be submitted.

Prerequisites

Concept	Video Lectures	in this lab
Dataflow modeling	CT Verilog Series 04	Wire, reg, signed, etc
Behavioral modeling	CT Verilog Series 05 ~ 06	Always block
Sequential blocks	CT Verilog Series 07	Synchronous output with clock

Complete these exercises before the Basic lab. They introduce the same clocked-design ideas with fewer edge cases.

1. Saturating Signed Adder with 1-Cycle Delay

Add two signed 8-bit two's-complement inputs, A and B. **If the mathematical sum is above 127, output 127. If it is below -128, output -128.**

1. Store out in a flip-flop. The result is updated at the next rising edge of clk (a 1-cycle delay).
2. **Update out only on the rising (positive) edge of clk.**
3. rst is synchronous and active-high. If rst is 1 at a rising edge, set out to 0; reset has priority.
4. Complete the SystemVerilog template lab2_1_practice.sv with the provided testbench and pattern file (lab2_1_practice_t.sv, pattern_A_practice.dat). Refer to the Appendix for instructions on adding the pattern files to simulation sources.
5. Run the provided simulation. A correct design displays PASS.

IO List and Specification:

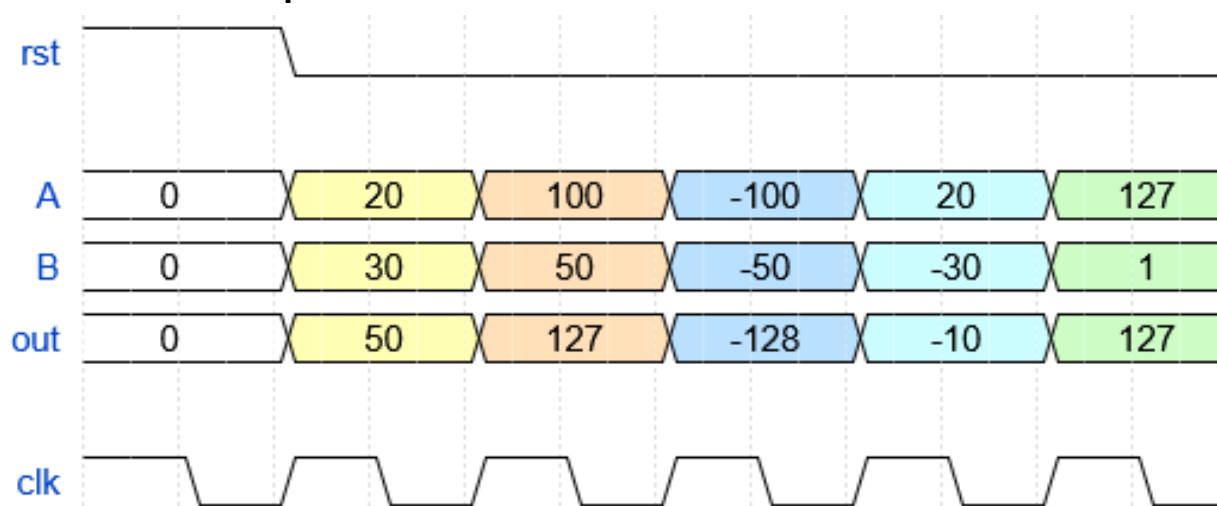
Signal	I/O	Bit Width	Description
clk	input	1	Clock (positive-edge triggered)
rst	input	1	Synchronous active-high reset
A	input	8	Signed operand
B	input	8	Signed operand
out	output	8	Signed saturated result (delayed by 1 cycle)

The testbench applies A and B at a falling edge. At the following rising edge, update out with their saturated sum. The testbench checks out 1 ns afterward.

Example

A	B	out
8'd100	8'd50	8'd127
-8'd100	-8'd50	-8'd128
8'd20	-8'd30	-8'd10
8'd127	8'd1	8'd127
-8'd128	-8'd1	-8'd128

Waveform example:



Display A, B, and out as signed decimal in the waveform.

Notes

- **Do not modify the testbench or patterns.**
- The provided Pass/Fail messages are only intended to assist debugging.
- **Use a 9-bit signed intermediate sum.** Otherwise, an 8-bit sum can wrap before saturation is detected.

2. Basic Countdown Timer

Design an unsigned 8-bit countdown timer. When load is 1 at a rising edge, capture N and start counting down.

1. After the rising edge with load at 1, out shows N. At each later rising edge, subtract 1 while out is greater than 0. At the edge that changes out from 1 to 0, set done to 1 for exactly one cycle.
2. Update out only on the rising (positive) edge of clk.
3. rst is synchronous and active-high. If rst is 1 at a rising edge, set out and done to 0; reset has priority.
4. Complete the SystemVerilog template lab2_2_practice.sv with the provided testbench and pattern files (lab2_2_practice_t.sv, pattern_B_practice.dat). Refer to the Appendix for instructions on adding the pattern files to simulation sources.
5. Run the provided simulation. A correct design displays PASS.

IO List and Specification:

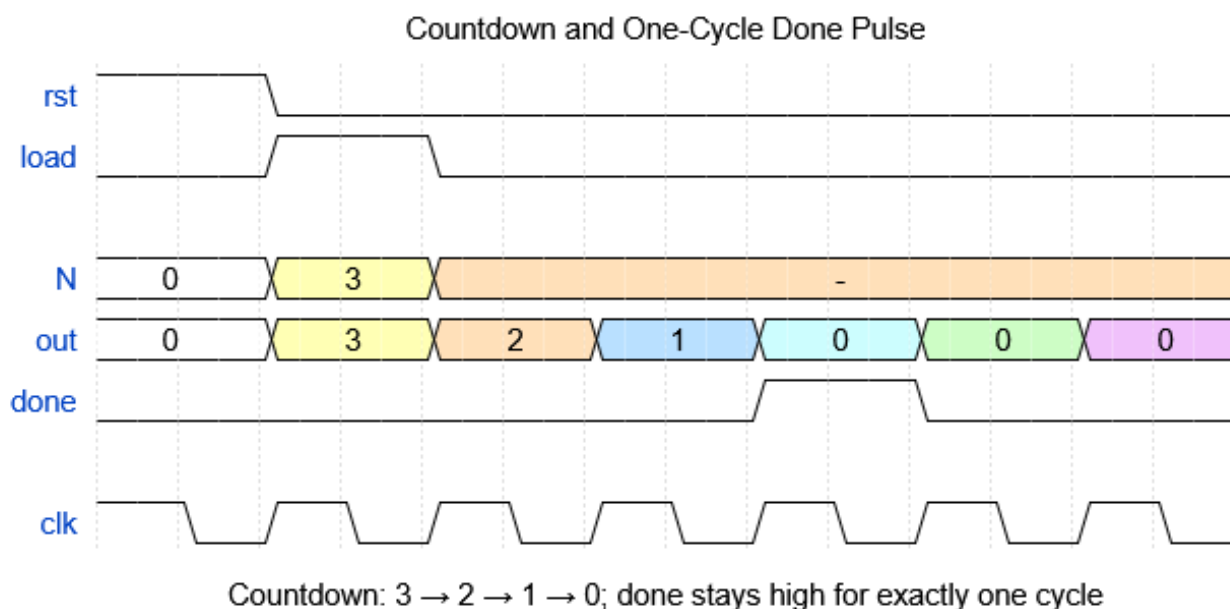
Signal	I/O	Bit Width	Description
clk	input	1	Clock (positive-edge triggered)
rst	input	1	Synchronous active-high reset
load	input	1	Captures N and starts the countdown
N	input	8	Unsigned initial value
out	output	8	Current counter value
done	output	1	One-cycle pulse when the count changes from 1 to 0

This practice uses one load per countdown and N is always greater than 0. The Basic lab adds reload and N = 0 cases.

Example

load	N	out	done
1	8'd3	8'd3	0
0	-	8'd2	0
0	-	8'd1	0
0	-	8'd0	1

Waveform example:



Display N and out as unsigned decimal; display load and done as binary.

Notes

- **Do not modify the provided testbench or patterns.**
- The provided Pass/Fail messages are only intended to assist debugging.
- Set done to 0 on every other cycle so it cannot remain high after the counter reaches zero.

Appendix: Adding Pattern Files to Vivado Simulation

1. In Add Sources, choose Add or create simulation sources, then select Add Files.
2. For Part 1, add pattern_A_practice.dat. For Part 2, add pattern_B_practice.dat.
3. In the file picker, change Files of type to All Files if the .dat files are not visible. Keep Copy sources into project selected so Vivado can find the pattern file during simulation.